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St. Joseph's College of Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore – 607001

CH305 - INORGANIC CHEMISTRY-II

Time : 3 hrs **Max Marks :75**

SECTION-A (20 x 1 = 20)

Answer ALL the questions

I. Choose the correct answer:

1. Which compound is probably most polar of the following?

a) boron trichloride b) oxygen difluoride

c) silicon tetrafluoride d) selenium difluoride

2. Among the following, which can be used to estimate Ni^{+2}

a.DMG b. OH^- c. Cl^- d. S^{-2}

3. The mineral, quartz, is an example of:

a) a single-chain silicate b) a framework silicate

c) a sheet silicate d) a double-chain silicate

4. Among the following, which is used as abrasive?

a) ZrC b) Be_2C c) Al_4C_3 d) SiC

5. Which one of the following atoms has two unpaired electrons in its ground state:

a) Sodium b) Calcium c) Oxygen d) Aluminum

6. The shape of SO_4^{-2} is

a) linear b) trigonal c. tetrahedral d. pyramidal

7. Rutile is the ore of

a) Ti b) V c) W d) Cr

8. Of these, the metal which occurs in a free state is _____.

a) Na b) Mg c) Zn d) Pt

9. The colour of Cr^{+3} is

a. purple b. green c. violet d. blue

10. The fertilizer which has highest nitrogen content is

a. Nitrolime stone b. Sindri fertilizer

c. Urea d. Nangal fertilizer

II. Fill up the blanks:

11. Elements in the Noble Gas family are considered stable because they have _____ outer energy levels.

12. . Clay minerals are common examples of _____ silicate structures.

13. Hydrazine is used as -------- fuel.

14. ozone is used to --------- air.

15. NiS ore can be concentrated by ---------- method.

III Match the following:

16. clay minerals - Van-Arkel process
17. olivine - product of chemical weathering (phase change), an important component of soil
18. Quartz - froth floatation
19. Sulphurores - product of lower-temperature igneous rock crystallization, or metamorphism, and is resistant to weathering so is found in many sedimentary rocks
20. Titanium - product of higher temperature igneous rock crystallization

SECTION-B (10 x2 = 20)

Answer any **TEN** questions:

21. Compare and contrast one significant feature of CCl_4&$SiCl_4$

22. Melting point of o-nitro phenol is 214^oC while those of its m- and p-isomers are 280^oC and 279 ^oC respectively. Why?

23. Calculate the oxidation number of Mn in $KMnO_4$.

24. Write short note on amphiboles.

25. Draw the structure of PCl_5

26. Mention any two uses of Noble gases.

27. Write any two exceptional properties of Fluorine.

28. How do you prepare Caro's acid?

29. Write short note on gravity separation method.

30. what are Psuedo halogens ? Give two examples.

31. How is pure tungsten metal prepared?

32. . Give the important oxidation states of chromium

SECTION-C (5 x7 = 35)

Answer any **FIVE** questions:

33.a. Explain the chemistry of ozone.. (4)

b. How are the metal atoms in a metal crystal bonded together? Explain.(3)

34.a. Balance the following equation by using ion-electron method: (4)

$Cr_2O_7^{2-} + Fe^{2+} + H^+ \rightarrow Cr^{3+} + Fe^{3+} + H2O$

b.Discuss the structure of chain silicates. (3 marks)

35. .i) Explain the chemistry of ozone. (3)

ii) Explain the hybridization and geometry of of XeF_4&XeF_6 (2+2)

36 .a. Explain the extraction of Titanium from its ore. (5)

b. Compare the properties of Zr and Hf. (2)

37. a. Compare H_2O and HF. (3)

b. How are peroxy acids of sulphur structurally different from thionic acids. (2)

c. Mention any two anomalous properties of oxygen. (2)

38. a. Discuss the position of inert gases in the periodic table. (4)

b. Write a note on HF as a solvent. (3)

39. a. Arrive at the structures of BrF_3, $C1F_5$ (4)

b. Explain the zone refining process. (3)

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